

Architecture Justification

Project: Telemedicine Slot Booking & Prescription Portal

Selected Architecture: Microservices Architecture

Reason 1 — Scalability

The system can experience different workloads across its functions. For example, appointment-slot booking may receive a large number of requests at the same time, while prescription services may have a different workload. With a microservices architecture, individual services can be scaled independently according to their demand without scaling the entire application.

Reason 2 — Fault Isolation

Microservices provide better fault isolation by separating the system into independent services. For example, if the Prescription Service experiences a failure, the Doctor & Slot Management Service and Consultation Service can continue operating independently. This reduces the effect of failures on the overall system.

Security Advantage

The system handles sensitive information such as patient details, consultation information, and digitally signed prescriptions. With microservices, security-sensitive functionality can be isolated into dedicated services with controlled interfaces. This allows access and security controls to be applied separately to different parts of the system.

Performance Benefit

The Consultation Service can be independently scaled to handle video consultation requests, while the Doctor & Slot Management Service can be scaled separately when appointment booking activity increases. This helps the system maintain good responsiveness under varying workloads.

Conclusion

Microservices Architecture is therefore suitable for the Telemedicine Slot Booking & Prescription Portal because it provides independent scalability, fault isolation, controlled access to sensitive functionality, and better performance under varying workloads.

Components:

Patient Management Service, Doctor & Slot Management Service, Consultation Service, Prescription Service, Doctor Service, Database

Interfaces:

Doctor & Specialty API, Slot Management API, Consultation API, Video Consultation API, Prescription API, Prescription Download API